Medical Record Science I — BBAHM 202

Complete Summary Notes | All 4 Modules Covered | 🌙 Dark Edition

Subject: Medical Record Science I | **Code:** BBAHM 202 (Major) | **Credits:** 5 (4T+1P)

University: Maulana Abul Kalam Azad University of Technology, West Bengal (MAKAUT)

Program: BBA in Hospital Management | **Mode:** Offline

Semester: 2nd Semester

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MODULE M1 (CO1)

Medical Records Basics

1.1 Historical Background

> **In easy words:** Hospitals have been writing down patient information for hundreds of years. India was slow to take this seriously, and it took several government committees to push for proper medical record departments.

- First medical record unit: **1667, St. Bartholomew's Hospital, England.**
- Pennsylvania Hospital, USA began a patient register in **1792.**
- Standardized inpatient records promoted by American College of Physicians & American College of Surgeons (late 20th century).
- In India: **Bhore Committee (1946)** first stressed importance of medical records; reiterated by **Mudaliar Committee (1962).**
- Jain Committee and Rao Committee highlighted poor state of records and recommended establishing a proper Medical Records Section in every hospital.

Example: *A 1960s district hospital in India had no organized record room — patient papers were just stacked in a cupboard. After the Mudaliar Committee's recommendations, the hospital set up a dedicated Medical Records Department with a filing system and a Medical Record Officer.*

1.2 Definition of Medical Records

> **In easy words:** A medical record is simply the written story of everything that happened to a patient in the hospital — what was wrong, what tests were done, and what treatment was given. It is also a legal paper, so it must be accurate.

- A medical record is an **orderly, written document** encompassing the patient's health history, physical examination, findings, laboratory reports, treatment, surgical procedure reports, and hospital course.
- It is a **legal document** providing a chronicle of a patient's medical history and care.
- When complete, it should contain sufficient data to justify investigations, diagnosis, treatment, length of stay, results of care, and future course of action.
- Physicians, nurse practitioners, and other members of the healthcare team may make entries.

1.3 Role of Medical Records in Healthcare Delivery

> **In easy words:** The same record file is useful to many different people — the patient, the doctor, the hospital, researchers, and even courts of law — each for a different reason.

For the Patient

- Documents clinical history
- Avoids unnecessary repeat tests/treatment
- Assists continuity of care
- Needed for insurance, employment, health schemes

For the Doctor

- Assures quality/adequacy of diagnosis & treatment
- Aids orderly continuity of care
- Supports evaluation of medical practice
- Aids research, continuing education
- Protects in legal matters

For the Hospital

- Documents type/quantity of work done
- Evaluates proficiency of doctors
- Evaluates services against norms
- Protects hospital legally
- Supports staffing/budget/planning decisions

For Education, Research & Legal Purposes

- Basis for all clinical research
- Furthers education of health personnel
- Supplies data to public health authorities
- Serves as legal evidence; vital for reimbursement and insurance

Example: *When a patient is readmitted a year later with the same chronic illness, the doctor pulls the old record instead of repeating every test from scratch — saving time, money, and avoiding unnecessary radiation/blood draws.*

1.4 Importance of Medical Records

> **In easy words:** Good records keep the patient's care continuous, protect everyone legally, and help the hospital run and plan better.

- Continuity of patient care
- Legal evidence and protection
- Research and education
- Billing and financial reimbursement
- Quality assurance
- Statistical data collection
- Administrative decision-making

1.5 Types of Medical Records

> **In easy words:** Not every record looks the same — an outpatient's slip is different from an admitted patient's full file, and accident or legal cases get special treatment.

Type	Description
Outpatient (OPD)	Records for patients attending the outpatient department; includes OPD card (9"×6"), identification data, history, physical findings, diagnosis, treatment, follow-up.
Inpatient (IPD)	Records for admitted patients; initiated at the admitting office with an admission number; includes all clinical documents from admission to discharge.
Emergency (Casualty)	Records for accident & emergency cases; includes registration, casualty number, medico-legal registration if applicable.
Diagnostic	Laboratory, radiology, pathology, and other investigation reports.
Surgical / Operation	Anaesthesia record, operation record, post-operative notes.
Medico-Legal (MLC)	Special records for cases with legal implications — accidents, assault, poisoning, etc.
Administrative	Census reports, statistical reports, personnel records.

1.6 Format Types of Medical Records

> **In easy words:** There are two main ways to organize what's written in a chart — by who wrote it (SOMR), or by the patient's problems (POMR).

SOMR
Source-Oriented Medical Record. Documents organized by source (physician notes, nursing notes, lab, radiology). Traditional method.

POMR
Problem-Oriented Medical Record. Documents organized around patient's problems. Introduced by Dr. Lawrence Weed. Uses SOAP notes. (Covered in detail in Module 2.)

1.7 Characteristics of a Good Medical Record

> **In easy words:** A good record is like a good witness statement — complete, true, readable, on time, private, easy to find, and follows the same format every time.

Characteristic	Meaning
Complete	Sufficient data to identify the patient; justify diagnosis, treatment, follow-up, and outcome.
Adequate	All necessary forms; all clinical information present.
Accurate	Capable of quantitative analysis; factually correct.
Legible	Readable by all authorized personnel.
Timely	Entered promptly at the time of care.
Confidential	Protected from unauthorized access.
Retrievable	Easy to locate and access when needed.
Standardized	Follows uniform formats and conventions.

1.8 Ownership of Medical Records

> **In easy words:** The hospital owns the physical file, but the patient owns the right to see and use the information inside it.

- Medical records are the **property of the hospital/healthcare facility** that created them.
- However, the **patient has the right to access** the information within their own records.
- The hospital is the custodian; the patient holds the right to information.
- Records cannot be released to third parties without the patient's written consent, except by court order or public health mandate.

1.9 Contents of Medical Records

> **In easy words:** A complete chart is a stack of specific forms, always in the same order, from the moment the patient is admitted to the moment they leave.

- Front sheet / Identification summary sheet (face sheet)
- Consent for treatment
- Legal documents (referral letter, request for information, etc.)
- Admission notes
- History and physical examination findings
- Clinical progress notes (doctor's and nurse's)
- Orders for treatment and medication forms
- Laboratory and investigation reports (X-ray, pathology, etc.)
- Anaesthesia record (if surgery performed)
- Operation report
- Temperature chart
- Discharge summary / referral slip

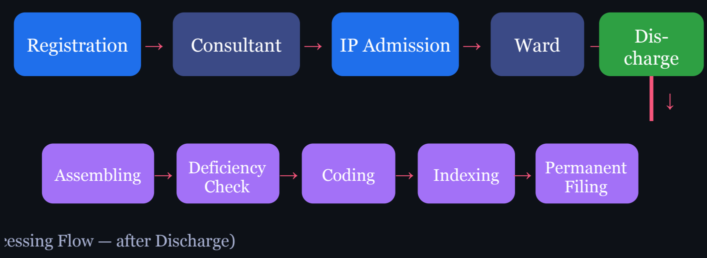
1.10 Maintenance of Records in the Ward

> **In easy words:** While the patient is on the ward, the nurse or ward clerk keeps the file up to date — admissions, discharges, reports coming in, and the daily head-count.

- Done by the ward nurse or ward clerk (AMRT — Assistant Medical Records Technician).
- Functions include: registration of admissions and discharges; receipt and mounting of investigation reports; maintenance of bed position accounts; scheduling follow-up appointments; preparation and submission of daily ward census reports.
- The admission check desk: receives admission advice; creates or updates patient index card; sends index card to incomplete record control desk; sends previous records to nursing unit; makes entries in the accession register.
- The census desk: prepares admission and discharge lists; collects discharged patient records from nursing units daily; prepares census reports.

1.11 Flow Chart of Functions (Inpatient Medical Record)

In easy words: A patient's journey through the hospital and a patient's file's journey through the Medical Records Department are two parallel flows — one is the person, the other is the paperwork.



Picture 1.1 — Patient flow (top) leads into the MRD processing flow (bottom) once the patient is discharged

Patient Flow

Registration Counter → Consultant → IP Admission → Ward → Discharge

MRD Processing Flow

Assembling → Deficiency Check → Coding → Indexing → Computer Entry → Scanning → Permanent Filing → Retrieval

Detailed Inpatient Flow

Nursing Unit / OPD / EMG / Labour / Newborn → On Discharge → Checking Date, Time, Diagnosis → Assembling → Checking → Coding → Indexing → Analysis → Reporting → Filing → Retrieval

(Incomplete records branch off "Checking" → sent to Doctors for completion → rejoin at Coding)

Example: *Mrs. Sharma is registered, seen by the consultant, admitted, treated on the ward, and discharged on Day 5. Her file then leaves the ward and enters the MRD processing flow — assembled into standard order, checked for missing signatures, coded with the ICD diagnosis, indexed, and filed — ready for retrieval if she returns or if her insurance company requests it.*

1.12 Assembling and Deficiency Check

In easy words: After discharge, all the loose papers of a patient's file are put together in one fixed order, then checked for anything missing — like a missing signature or test report — before they are sent away to be properly filed.

Standard Assembly Order:

1. Face sheet (identification/summary sheet)
2. Case summary and discharge records
3. History and physical examination
4. Labour record (if applicable)
5. Consultation record
6. Clinical progress notes
7. Laboratory and other investigation reports
8. Anaesthesia record
9. Operation record
10. Nurses' records
11. Temperature chart
12. OPD card

Deficiency Check Process:

- Collect all loose documents after discharge and arrange in standard order.
- Tag case sheets; ensure MR number is present on all sheets.
- Check each record for: missing signatures, final diagnoses, investigation reports, consent forms.
- Complete case sheets → filed in permanent filing area.
- Incomplete case sheets → deficiency slip → sent to doctors' room for completion.
- Incomplete records filed unit-wise until completed.
- List of incomplete records submitted to hospital administration periodically.

Example: *On checking a discharged patient's file, the MRT notices the operation note has no consultant signature. A deficiency slip is attached and the file is routed to the Doctors' Conference Room for the surgeon to sign before it can be permanently filed.*

MODULE M2 (CO2)

Coding and Record Management

2.1 Medical Coding — Definition and Purpose

> **In easy words:** Coding means turning a diagnosis or procedure written in plain words into a short universal code, like translating "broken arm" into a number every hospital in the world understands.

- Medical coding is the process of **translating diagnoses, procedures, and supply information** from patient records into universal alphanumeric codes.
- These codes are used by billers to submit claims to insurance companies.
- Used for: billing, statistics, research, insurance reimbursement, public health reporting.
- In each medical record, an international code number is assigned to the diagnosis based on ICD (International Classification of Diseases) by WHO.
- Medical coders must be knowledgeable in medical terminology and anatomy.

2.2 Types of Medical Codes

Code System	Purpose
ICD codes (ICD-9, ICD-10, ICD-11)	Describe illness, injury, or cause of death
CPT (Current Procedural Terminology)	Surgeries, radiology, anaesthesia, measurement procedures
HCPCS Level 2	Outpatient hospital care and medical aid

2.3 Six Key Steps in the Medical Coding Process

> **In easy words:** A coder reads the doctor's notes carefully, asks questions if anything is unclear, picks the most accurate codes, makes sure the codes match each other, and double-checks everything before submitting.

1. Abstract Documentation → 2. Query if Necessary → 3. Code Diagnosis → 4. Code Procedures → 5. Confirm Medical Necessity → 6. Double-Check Codes

1. **Abstract the documentation:** Read all physician notes completely — procedure notes, operative reports, history & physical. Understand Who, Where, Why, What, How. Accuracy is paramount.
2. **Query if necessary:** Ask for clarification if documentation is unclear. "If it's not documented, it didn't happen. If it didn't happen, you can't code it." Legal queries must ask open-ended questions.
3. **Code the diagnosis/diagnoses:** Select the most accurate, most specific code. Review inclusive signs and symptoms.
4. **Code the procedure/procedures:** Assign codes for all procedures. Note inclusive procedural components (e.g., simple repair after excision is included in the Excision code).
5. **Confirm medical necessity:** Align diagnosis codes with procedure codes to justify the provision of treatment. Essential for reimbursement and legal/ethical review.
6. **Double-check your codes:** Code backwards — look up the code, re-read the description, and compare with documentation.

Example: *A coder reading an operative report for "laparoscopic cholecystectomy for symptomatic gallstones" assigns the ICD diagnosis code for cholelithiasis and the CPT procedure code for laparoscopic cholecystectomy, then checks that the diagnosis justifies the procedure before submitting the claim.*

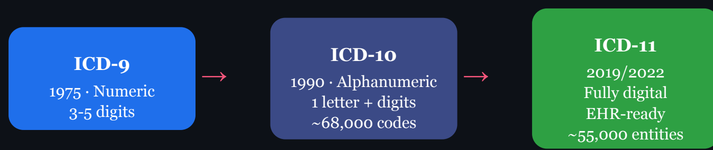
2.4 ICD — International Classification of Diseases

> **In easy words:** ICD is WHO's worldwide dictionary of disease codes. It has been updated three times — ICD-9, ICD-10, and now ICD-11 — each version more detailed and more digital-friendly than the last.

A **standardized diagnostic classification system** by WHO that assigns unique codes to diseases, signs and symptoms, abnormal findings, injuries, external causes of injury, and factors influencing health status.

Objectives of ICD:

1. Standardization of health information worldwide.
2. Disease surveillance — monitoring occurrence and spread of diseases.
3. Mortality and morbidity statistics.
4. Health planning and policy development.
5. Research and epidemiology.
6. Health insurance and reimbursement.



Picture 2.1 — Evolution of ICD across the three revisions, growing more detailed and more digital each time

ICD-9 (9th Revision — 1975):

- Consists mainly of **numeric codes** (3–5 digits).
- Used for recording morbidity and mortality statistics.
- Widely used in hospitals and public health agencies.
- **Limitations:** Limited number of codes; less detailed; difficult to accommodate new diseases; not suited for modern EHR systems.
- Examples: 250.0 = Diabetes Mellitus; 401.9 = Essential Hypertension; 486 = Pneumonia.

ICD-10 (10th Revision — Introduced 1990):

- Uses **alphanumeric codes** (one letter followed by numbers).
- Much larger number of codes; greater specificity and detail.
- Covers diseases, disorders, injuries, causes of death, factors affecting health.
- Supports electronic health records and computerized health information systems.
- **Advantages:** More accurate classification; better tracking; improved international comparability; accommodates new diseases; supports insurance and research.
- Examples: E11 = Type 2 Diabetes Mellitus; I10 = Essential Hypertension; J18.9 = Pneumonia, Unspecified.

ICD-11 (11th Revision — Adopted 2019; effective globally 1 January 2022):

- Designed specifically for **digital and electronic health systems.**
- More detailed and comprehensive classification; includes newly recognized diseases.
- Fully digital and user-friendly; easy to search, code, and update.
- Integrates easily with hospital information systems and EHR.
- Improves international comparison of health data.
- Example: 5A11 = Type 2 Diabetes Mellitus.

Feature	ICD-9	ICD-10	ICD-11
Year Introduced	1975	1990	2019 (effective 2022)
Code Format	Numeric (3–5 digits)	Alphanumeric (1 letter + digits)	Alphanumeric (fully digital)
Number of Codes	~13,000	~68,000+	~55,000 entities (expandable)
EHR Compatible	No	Partially	Yes (fully digital)
Specificity	Low	High	Very High

Example: *Under ICD-9 a diabetic patient was coded simply "250.0". Under ICD-10 the same patient is coded "E11" with room for more detail (e.g., complications). Under ICD-11, the digital system "5A11" links directly into the hospital's electronic record and updates automatically with new clinical knowledge.*

2.5 POMR — Problem-Oriented Medical Record

In easy words: Instead of writing notes department by department, POMR organizes the whole chart around the patient's list of problems, and every day's note for each problem follows the same SOAP pattern.

- Introduced by **Dr. Lawrence Weed, MD.**
- A structured method of recording patient medical history focused on the patient's **problems** and their clinical evolution.
- Allows physicians to view the patient's history in a systematic and organized manner.

Picture 2.2 — The 4 phases of POMR, from gathering data to reviewing how well the plan worked

4 Phases of POMR:

1. Formation of a **Database** (history, physical examination, lab and test results).
2. Identification of a **Problem List** from the database (active and inactive problems).
3. Identification of a specific **Plan of Care (POC)** for each problem; includes evaluative and progress notes.
4. Determination of the **effectiveness of the POC** and changes based on patient progress.

5 Basic Components of POMR:

1. **Database:** Complete history, physical exam, and laboratory data. Additional information added as collected.
2. **Complete Problem List:** Constructed after database; problems are either active or inactive. Inactive = prior resolved illnesses still important to remember.
3. **Initial Plans:** Written by admitting physician after problem list; states "what to do about what is wrong."
4. **Daily Progress Notes:** Written in SOAP format for each active problem. A note need not be written every day if nothing has changed.
5. **Final Progress Note / Discharge Summary:** All active problems defined to their furthest resolution; includes S, O, A, P for each; emphasis on unresolved problems.

SOAP Format (Progress Notes):

S — Subjective → O — Objective → A — Assessment → P — Plan

Letter	Meaning	Description
S	Subjective	Brief review of abnormalities identified in history pertaining to each problem; what the patient reports.
O	Objective	Brief review of physical examination and initial lab data; measurable findings.
A	Assessment	Physician's interpretation of the problem; differential diagnosis in prioritized manner if needed.
P	Plan	Three distinct groupings: 1. Diagnostic Plan — workup to rule in/out each differential. 2. Therapeutic Plan — initial therapies and rationale. 3. Patient Education Plan — plans to educate patient.

Advantages of POMR

- Better organization of patient information
- Improved communication among care team
- Better coordination of care; improved outcomes
- Team collaborates on problem list and plan of care

Disadvantages of POMR

- Too complex to maintain after initial entries
- Several problems discussed in one encounter — cumbersome
- Inhibits data synthesis; lengthy progress notes

Example (SOAP note): *Problem — Type 2 Diabetes.* ***S:*** *patient reports increased thirst and fatigue.* ***O:*** *random blood glucose 240 mg/dL, BP 130/85.* ***A:*** *poorly controlled Type 2 DM.* ***P:*** *increase metformin dose (therapeutic), order HbA1c (diagnostic), counsel on diet and exercise (patient education).*

2.6 Indexing

> **In easy words:** Indexing is like making a library catalogue for patient files — separate lists by patient name, by disease, by operation, or by doctor — so any file can be found quickly even without the MR number.

Indexing is the process of maintaining or arranging patient medical reports in a systematic order (chronological, alphabetical, numerical, by specialty, by physician, etc.) to **locate specific documents quickly.**

Types of Indexes:

Index Type	Description
Alphabetic / Master Patient Index (MPI)	Indexed by patient's name in alphabetical order. Provides entry into the filing system. Unique identifier for each patient. Used when the MR number is not known.
Disease Index	Catalogue (3"×5" or 5"×8" cards) of clinical records of patients with same diagnosis, indexed by ICD code. Includes patient identification, age, gender, result of treatment, complications.
Operation / Procedure Index	Catalogue of patients who have undergone operations; indexed by surgical/medical procedures.
Physician Index	Catalogue of all patients treated by a particular physician; used to evaluate physician performance.
Unit Index	Details of patients treated in a particular unit; used to evaluate unit performance.

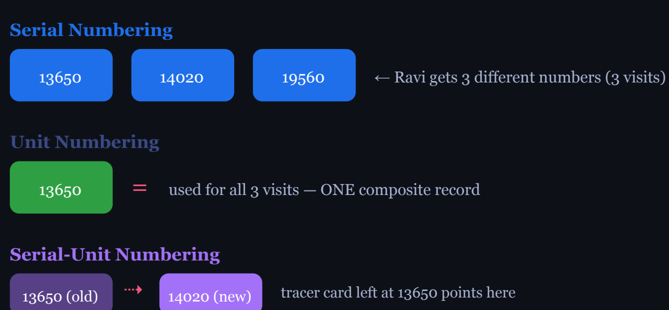
Purposes of Indexing:

- Review cases of disease for current health problem management.
- Compose data for scientific papers.
- Procure data on utilization of hospital facilities.
- Evaluate quality of care.
- Provide data for hospital committees and drug trials.

Example: *A researcher studying outcomes of appendectomies pulls every record from the Operation Index coded under "appendectomy" over the last 5 years, instead of searching the entire MRD record by record.*

2.7 Medical Record Numbering Systems

> **In easy words:** A hospital must decide whether the same patient gets a brand-new number every visit, or keeps just one number forever. Each method has trade-offs between simplicity and having one combined file per patient.



Picture 2.3 — Serial vs. Unit vs. Serial-Unit numbering: how many numbers the same patient ends up with

System	Description	Example
Serial Numbering	Patient receives a new number each time they register. Multiple numbers for same patient.	Ravi: 1st visit = 13650; 2nd visit = 14020; 3rd visit = 19560
Unit Numbering	Patient assigned one number on first visit; same number used for all subsequent visits. One composite record.	Ravi always uses number 13650
Serial Unit Numbering	New number each visit, but old records brought forward and filed under the latest number. Tracer card left in old location.	Ravi gets 14020 on 2nd visit; old record 13650 filed with it; tracer card at 13650 points to 14020
Annual Numbering	Two digits indicating the year added to end of serial number. Controls inactivation of records.	—
Family Numbering	Extra two digits added to indicate individual in a family: 01=Head; 02=Spouse; 03=Children; 04=Other relatives.	—

Example: *Under Unit Numbering, Ravi's blood tests, X-rays, and discharge summaries from every visit over 10 years all sit inside the single folder numbered 13650 — giving the doctor his whole history at a glance.*

2.8 Filing

> **In easy words:** Filing means deciding exactly where each patient's folder physically sits on the shelf, and using their MR number (read in a clever order) to make sure new files don't all pile up in one place.

Systems of Filing:

System	Description	Advantage	Disadvantage
Centralized Filing	All records filed centrally in the MRD.	More efficient; better control; cost-effective. Preferred by most hospitals.	—
Decentralized Filing	Each department keeps its own records independently.	Convenient for each department.	Labour intensive; higher operating costs; duplication of effort.

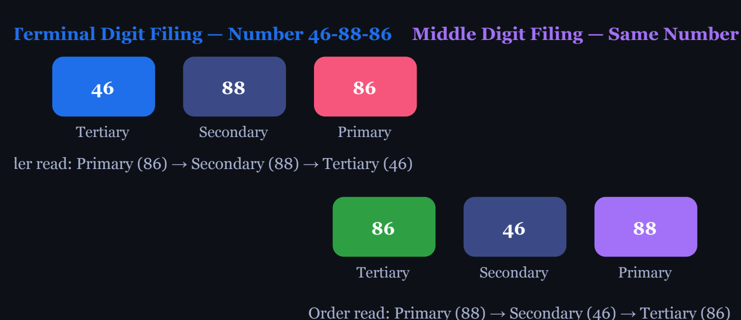
Methods of Filing (Numeric):

1. Straight / Serial Numeric Filing:

- Records filed in exact ascending order of registration number (e.g., 65023, 65024, 65025...).
- **Advantage:** Easiest method; simplest to train staff.
- **Disadvantages:** Easy to misfile (all digits must be considered at once); heaviest filing activity concentrated at one end (most recent records); equal distribution of work is difficult.

2. Terminal Digit Filing & 3. Middle Digit Filing:

> **In easy words:** Instead of reading a 6-digit number left to right, the clerk reads it in pairs in a different order — this spreads new files evenly across 100 shelves instead of jamming them all at the end.



Picture 2.4 — Same 6-digit number 46-88-86 split differently: Terminal Digit (right pair first) vs. Middle Digit (middle pair first)

- Six-digit number divided into three pairs: **Primary digits = last two (right); Secondary = middle two; Tertiary = first two (left).**

- 100 primary sections (00–99). Clerk first goes to the primary section, then secondary, then files by tertiary digits in order.
- Example: Number 46-88-86 → go to section 86, find subsection 88, file by 46.
- **Advantages:** Even distribution of new records across 100 sections; congestion eliminated; misfiling substantially reduced (clerk focuses on only one pair of digits at a time); inactive records can be pulled section by section as new records arrive; clerks assigned responsibility for certain sections.
- **Disadvantage:** Slightly longer training period.

3. Middle Digit Filing:

- **Primary digits = middle two; Secondary = left two; Tertiary = right two.**
- Example: 46 (secondary) – 88 (primary) – 86 (tertiary).
- **Advantages:** Simple to pull 100 consecutively numbered records for research; easier conversion from straight numerical than terminal digit; better distribution than straight numerical; misfiling reduced.
- **Disadvantage:** More training required than straight numeric.

4. Alphabetical Filing:

- Records filed by patient name in alphabetical order.
- Suitable for small hospitals with low patient volumes.
- **Disadvantage:** Difficult to manage in large hospitals; risk of filing errors with common names.

Example: *A clerk asked to file record number 46-88-86 under Terminal Digit Filing walks straight to shelf-section "86," finds the "88" subsection, and slots it in numerically by "46" — without ever needing to scan the whole 6-digit number at once.*

Physical Types of Filing:

- Vertical filing
- Suspended (lateral) filing
- Horizontal filing

Colour Coding of Folders:

- Colour bars on folder edges create distinct patterns in different sections of the file.
- A break in the colour pattern signals a misfiled record.
- Most effective with terminal digit or middle digit filing.
- One approach: 10 colours for primary digits 0–9. Two colour bars signify each of the two primary digits.

2.9 Microfilming

> **In easy words:** Microfilming shrinks paper records onto tiny film so a whole record room can fit into a small cabinet — saving space, but slow to update.

- Microcopy/microfilm: Making microscopic copies of documents for preservation and space saving.
- Used in large teaching hospitals where space is constrained.
- Saves space by **up to 90%.**

Types of Microfilm:

Type	Description
Roll Film	Sits on a reel; comes in 16 mm and 35 mm formats.
Microfiche — COM	Images in a grid; can hold up to 300 pages.
Microfiche — Jacket	Used for camera film; placed in plastic jackets. Usually 5 images across.
Aperture Cards	Timecard-like with a square hole for film; holds one image; used in engineering.

Advantages

- Massive space saving (90%)
- Easy accessibility and retrieval
- Protection of original records
- Elimination of misfiling
- Durability — can last hundreds of years if stored properly
- Lower cost than computers/hard drives
- Saving of time and manpower

Disadvantages

- Slow retrieval rate; very hard to update
- Only one person can view images from a roll at a time
- Deteriorates with heat, humidity, light, or chemicals

Storage Conditions for Microfilm:

- Sealed containers in metal cabinets treated with non-corrosive, non-staining, non-combustible paint.
- Avoid heat, humidity, light exposure, and proximity to chemicals.
- Create duplicate backup copies or digitize the collection.

Equipment Required for Microfilming:

- Microfilming camera, processors, viewing machines, duplicating and xerox machines, microfilm rolls, fixer and developer, microfilming technicians.

2.10 Computerization of Medical Records

> **In easy words:** Storing records on computers instead of paper lets many authorized staff see the same record at once, from anywhere, while still being kept private.

- Possible to store text and all types of images (X-rays, CT scans, MRI).
- By networking, access can be provided to doctors, nurses, technicians, and administrators while maintaining confidentiality.

EHR vs. EMR:

EMR (Electronic Medical Record)	EHR (Electronic Health Record)
Digital record from a single provider/clinic.	Accessible anywhere; can be shared across providers.
Not intended for distribution outside individual practice.	Enables real-time, patient-centred, collaborative care.
Stored locally.	Cloud-based or networked.

Benefits of EHR

- Accurate, up-to-date, complete information at point of care
- Quick access for coordinated, efficient care
- Reduces medical errors; supports safer prescribing
- Improves documentation, coding, and billing
- Enhances privacy and security
- Reduces costs (less paperwork, less duplicate testing)
- Enables remote access by authorized users

Disadvantages of EHR

- Outdated data if not updated immediately
- High setup costs and digitization time
- Cybersecurity risks; PHI breach risk
- Impersonal patient-doctor interaction
- Frustration with poor implementation

2.11 Hospital Statistics

> **In easy words:** Hospitals turn raw numbers — beds, discharges, deaths — into formulas that show how efficiently the hospital is running.

Key Statistical Formulas:

```
Bed Occupancy Rate (BOR) = (Occupied Bed Days / Available Bed Days) × 100
```

```
Average Length of Stay (ALOS) = Total Inpatient Days / Total Discharges (including deaths)
```

```
Gross Death Rate = (Total Deaths / Total Discharges including deaths) × 100
```

```
Net Death Rate = (Deaths occurring after 48 hours of admission / Total Discharges) × 100
```

```
Bed Turnover Rate = Total Discharges / Average Beds Available
```

```
Bed Turnover Interval = (Average Beds × Period) / Total Discharges
```

```
OPD/IPD Ratio = Total OPD Patients / Total IPD Admissions
```

Types of Reports:

- Daily, weekly, fortnightly, monthly, quarterly, half-yearly, and annual reports.
- Types: Vital statistics; ADT (Admission, Discharge, Transfer) analysis; General health statistics.

Uses of Hospital Statistics:

- Hospital planning and resource allocation.
- Evaluation of quality of care.
- Government reporting and compliance.
- Research and epidemiological studies.
- Locating deficiencies in staff, facilities, equipment, methods, and outcomes.
- Prevention of diseases.

Example: *A 200-bed hospital had 180 occupied bed-days out of 200 available bed-days in a day → BOR = (180/200) × 100 = 90%, indicating very high bed utilization.*

2.12 Process of Arranging Medical Records

> **In easy words:** This is the full pipeline a discharged patient's paperwork travels through before it can be retrieved again later.

Collect Census + Discharged Records → Analysis → Assembly + Deficiency Check → Coding + Indexing → Computer Entry + Scanning → Permanent Filing → Retrieval

- Collection of daily ward census reports along with discharged patient records.
- Analysis of discharged patient records.
- Assembly in prescribed standard order; deficiency check.
- Coding and indexing as per latest ICD revision.
- Computer entry and scanning.
- Permanent filing.
- Retrieval as needed.

Reasons Records Are Retrieved:

- Follow-up of patients.
- Admission to ward/casualty for observation.
- Research workers for academic purposes.
- Medical reimbursement claims.
- Production in court of law.

MODULE M3 (CO3)

Medical Record Department (MRD)

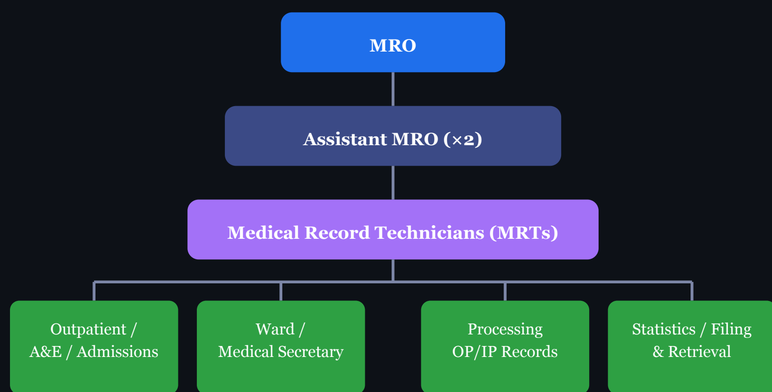
3.1 Goal of MRD

> **In easy words:** The MRD's job is to keep every patient's record accurate, complete, and modern enough to be useful to doctors, the hospital, and the law.

- To maintain a **modern and scientific** medical record system in conformity with international standards.
- To ensure that medical records contain **comprehensive and accurate social, medical, and vital information** to facilitate effective medical services.

3.2 Organization and Management of MRD

> **In easy words:** The MRD is run like a small company — one Medical Record Officer at the top, assistants below, technicians below them, and each technician's team handles a different department's records.



Picture 3.1 — MRD organizational structure: MRO at the top, down through Assistant MROs and MRTs, to the functional sections

Organizational Structure (listed):

Medical Record Officer (MRO) → Assistant MRO (×2) → Medical Record Technicians (MRTs) → Departments: Outpatient (CR & Appointment; OP Clinic) | A&E | Admissions | Ward | Medical Secretary | Processing OP Records | Processing IP Records | Statistics | Filing Records & X-rays | Retrieval Records & X-rays | General.

Four Functional Categories of MRD:

1. **Outpatient Services:**
   - A. Central Registration and Appointment (3 shifts, 24×7): control of hospital numbers; preparation of pre-numbered folders; registration of new cases; maintenance of Master Patient Index; registration of follow-up appointments; supply of records to OPD clinics.
   - B. Outpatient Clinic: collection of new and follow-up patient records including X-rays; maintenance of patient treatment accounts; collection and return of records to files.
2. **Accident & Emergency (Casualty) Service (3 shifts, 24×7):**
   - Control of casualty numbers; registration of A&E and MLC cases; collection and mounting of investigation reports; referral of patients; collection of statistics and filing of A&E records.
3. **Inpatient Service:**
   - A. Admissions Office (3 shifts): maintenance of waiting list; registration of admissions and discharges; maintenance of bed occupancy board; securing patient property.
   - B. Ward (ward nurse/AMRT): registration; mounting investigation reports; bed position accounts; scheduling follow-up appointments; daily census reports.
4. **Medical Record Library (3 shifts, 24×7):**
   - Processing of outpatient and inpatient records.
   - Collection and analysis of hospital statistics.
   - Filing and retrieval of records and X-rays.

- General medical record functions.

Example: *A patient arrives by ambulance at 2 a.m. — the A&E section (running 24×7) registers him under a casualty number; once admitted, the Admissions Office takes over with an IP number; on discharge, his file moves to the Medical Record Library for processing and filing.*

3.3 Staffing Guidelines for MRD

> **In easy words:** The bigger the hospital, the more MRD staff are needed — there's a simple formula based on working days, shifts, and bed count.

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Formula: (Working Days × Number of Shifts) / Man-Days = Number of Persons Required
Example: (365 × 3) / 217 = 5 persons approx. for one unit (Man-days = actual working days after subtracting holidays)
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Hospital Size	Personnel Formula	Result
100 beds	2 (base) + 1 per 100 beds + 1 per 10 beds	2 + 1 + 10 = 13 personnel
500 beds	2 + 5 + 50	57 personnel

> General guideline: minimum 2 persons for any size hospital, plus additional based on bed count. Factors affecting staffing: bed strength, daily patient load, type of hospital, degree of computerization.

3.4 Role of MRD Personnel

> **In easy words:** The MRO manages the whole department and its policies; the MRTs do the hands-on daily work — coding, filing, assembling, and indexing.

Duties and Responsibilities of Medical Record Officer (MRO):

- Establish, organize, and manage the MRD effectively.
- Develop policies and procedures as per the Health Directorate/Ministry of Health.
- Assist the Medical Record Committee in developing different forms.
- Review medical records of different departments.
- Cooperate with other departments' staff.
- Participate in quality assurance, utilization review, infection control, and other committees.
- Prepare medical reports, certificates, birth/death registers; notify concerned authorities.
- Prepare monthly statistical reports on hospital activities.
- Observe professional ethics and protect confidentiality of patient information.
- Participate in research programmes.
- Select appropriate personnel for MRD; prepare training programmes.
- Design and develop standard basic medical record forms.
- Maintain effective Master Patient Index.
- Carry out quantitative and qualitative analysis of records concurrently and retrospectively.
- Establish "Incomplete Medical Records Control" — review all discharged patient records promptly.
- Create "Doctors Conference Room" in MRD for doctors to complete deficient records.
- Submit regular reports on delinquent records with respective doctors' names.

Role of Medical Record Technician (MRT):

- Coding diagnoses using ICD.
- Assembling discharged patient records in standard order.
- Deficiency checking.
- Indexing records by disease, procedure, physician, and patient.
- Filing and retrieval of records and X-rays.
- Collection and compilation of hospital statistics.
- Medico-legal documentation assistance.

- Registration of new patients and maintenance of Master Patient Index.

3.5 Retention of Medical Records

> **In easy words:** "Retention period" means how long a record must legally be kept before it can be destroyed — death and birth records are kept forever, but most ordinary patient files are kept for 5-10 years.

Retention period is the duration for which information should be maintained, irrespective of format (paper, electronic, or other).

Factors Determining Retention Period:

- Internal organizational need.
- Regulatory requirements for inspection or audit.
- Legal statutes of limitation.
- Involvement in litigation.
- Taxation and financial reporting needs.
- Mandatory state statutes.

General Retention Guidelines:

Category	Retention Period
Adult patients (general)	5–10 years from date of last visit
Minors (children)	Until age of majority + specified years
Medico-Legal Cases (MLC)	10+ years (as per legal requirements)
Disease index cards	10 years
Operative index	10 years
Statistical data	5 years
Death records	Permanent
Birth records	Permanent

Important Reasons to Retain Records:

- To meet the needs of the patient.
- To protect the legal interests of the patient, attending staff, and hospital.
- To assist in teaching medical and paramedical staff.
- To carry out medical research.

Example: *A child treated for a fracture at age 8 will have that record retained until they turn 18 (age of majority) plus a further specified number of years — far longer than a routine adult OPD visit record.*

3.6 Preservation of Medical Records

> **In easy words:** Preservation means physically protecting paper records from rot, insects, fire, pollution, and bad climate, or keeping electronic copies safely backed up.

Manual Preservation Methods:

1. **Selection of Paper and Ink:** For records needing to last 10+ years, use specific ink and paper developed in USA and UK for long-term preservation.
2. **Protection from Decay and Rot:** Ensure proper cross-ventilation; adequate fans or air-conditioning; regular dusting with vacuum cleaners. Avoid hot, dry climate; stagnant air; direct sunlight; dust accumulation; dampness.
3. **Protection from Insect Attack:** Periodic checking of floors and walls; insecticidal powders/sprays; naphthalene balls in bookshelves; fumigation using non-damaging fumigants. Filing racks should be 5" away from damp walls. Use closed steel racks (wooden racks attract white ants).
4. **Atmospheric Pollution Control:** Records deteriorate faster in cities due to sulphur dioxide and sulfurated hydrogen. Dust acts as nucleus for acidic gas condensation. Use alkaline water treatment in air-conditioning (pH 8.6 or above) to neutralize acidic gases.

5. **Fire Safety:** No smoking, open flames, or chemical storage in record room. Adequate fire extinguishers at convenient places. All electric wires through conduits. Windows and ventilators covered with wire-net frames.
6. **Temperature and Humidity Control:**

   ```
   Ideal: Temperature 22–25°C | Relative Humidity 45–55%
   ```

   Temperature >32°C and humidity >70% causes growth of hookworms, silverfish, cockroaches, termites, and fungi. Use air-conditioning and commercial humidifiers.
7. **Microfilming Control Register:** Maintain a register with: hospital number, patient name, age, sex, date last admitted, date microfilmed, microfilm file number, date original record destroyed, and remarks.

Electronic Preservation Methods:

- Hospital Information System (HIS)
- Electronic Medical Records (EMR)
- Electronic Health Records (EHR)
- Picture Archiving and Communication System (PACS) — for radiology.
- Scanning and digitization of paper records.
- Cloud storage.
- RAID (Redundant Array of Independent Disks).
- Optical discs.
- Encryption for data security.
- Automated backup systems.

Destruction Procedure:

- Inactive records destroyed as per retention schedule.
- Plan includes: method of destruction (shredding or burning); place for destruction; mode of transportation from MRD.
- A **Destruction Register** must be maintained — records hospital number, patient name, age, sex, address, date of registration, admission status, final diagnosis, and operations before destruction.
- Written permission must be obtained from hospital administration before destruction; all departments to be informed.

Example: *A record room exceeding 32°C and 70% humidity during monsoon season begins to show silverfish damage on old files — prompting the MRO to install dehumidifiers and schedule urgent microfilming of the most at-risk records.*

Medical Record Administration & Legal Aspects

4.1 Medico-Legal Cases (MLC)

In easy words: An MLC is any injury or illness case where the police need to investigate to find out who caused it — like an accident, assault, or poisoning — and it must be documented very carefully because it can end up in court.

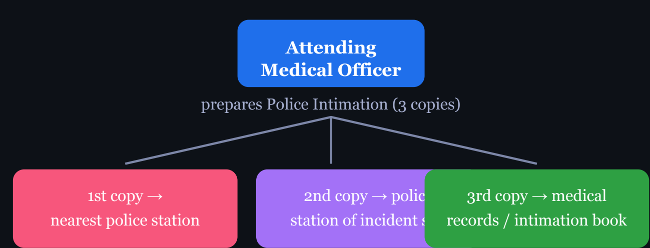
A MLC is a **case of injury or ailment in which investigation by law enforcement agencies is essential** to fix responsibility regarding the causation of the said injury or ailment. It involves both medical and legal aspects.

Incidents that Come Under MLC:

- Assault and battery, domestic violence, child abuse.
- Road Traffic Accidents (RTA), industrial accidents, cases of trauma.
- Electrical injuries, chemical injuries, burns and scalds.
- Poisoning, alcohol intoxication, drug overdose.
- Undiagnosed coma.
- Sexual offences, criminal abortions.
- Attempted suicide.
- Asphyxia (hanging, drowning, suffocation, strangulation).
- Custodial deaths, death in the operation theatre.
- Unnatural deaths, death due to snake or animal bite.
- Firearms injuries (bullet/stab injury).
- Dead brought to A&E (Found Dead); death within 24 hours of hospitalization without establishment of diagnosis.

Police Intimation (Under Section 39, CrPC):

In easy words: The doctor must tell the police about every MLC — this is the law, not optional — and a copy of the intimation note goes to three different places.



Picture 4.1 — One police intimation note, three copies, three destinations

- The attending Medical Officer is **legally bound** to inform the police about the arrival of an MLC.
- Failure to report may invite prosecution under Sections 176 and/or 202 of I.P.C.
- Information also given to ADH and Station Headquarters.
- 3 copies of police intimation: 1st copy → nearest police station; 2nd copy → police station of incident location; 3rd copy → medical records or retained in police intimation book.
- Police informed on discharge, transfer, or death of the patient.

Documentation of MLC:

- Done in duplicate in a set Proforma; clear, legible handwriting; complete words; no shortcuts or abbreviations.
- No cutting/overwriting; all corrections properly initialled.
- All columns filled by the same doctor who examined the patient.
- Completed then and there — no provisions left to complete later.
- Investigation findings, treatment given, and opinion to be recorded in MLC sheet.
- If opinion cannot be given, write "under observation" and sign.

- After all investigations, the same doctor gives the final opinion in the original MLC sheet.
- Doctor signs with full name and designation.
- After registration as MLC, all subsequent documents bear the same MLC number.
- All MLC X-rays kept in department as evidence for court.
- If patient is dead: handed over to police for post-mortem, not to relatives.
- Copy of MLC sheet handed over to police for investigation (against their signature and number).

MLC Form — General Details Recorded:

- Registration number; MLC number; Name; S/D/W of (Son/Daughter/Wife of); Age; Sex; Religion; Occupation; Residential address; Brought by; Date and time of examination; Name of police station.

Special Preservation of Evidence:

Case Type	Evidence to Preserve
Bullet/Stab/Burn Injury	Clothing with tears, holes, cuts, blood stains. Bullets recovered from body (marked by etching before preservation). Each piece of evidence circled and numbered with matching MLC description.
Poisoning	Gastric lavage/contents/vomitus, soiled clothing. Blood, urine, and relevant body fluids.
Sexual Offences	Clothing with blood stains, seminal stains, tears/cuts. Vaginal swab.

Transfer of MLCs:

- Transferring hospital provides treatment within its capacity and minimizes risk.
- Receiving hospital must have space, qualified personnel, and have agreed to accept transfer.
- All medical records (history, examination, diagnostic results, provisional diagnosis, treatment) transferred with the patient.
- Written informed consent/certification required.
- Transfer via qualified personnel with life support equipment.

Brought Dead Cases:

- All cases of unnatural death (accidents, burns, violence, sexual assault) labelled as medico-legal; police constable on duty informed.
- In death certificate: cause of death = **"NOT KNOWN BROUGHT DEAD."**
- Never write that cause of death should be determined after post-mortem in the death certificate.

Example: *A motorcycle accident victim is brought to A&E unconscious. The doctor registers it as an MLC, sends police intimation copies to the nearest station and the station covering the accident site, and documents in the MLC proforma that the patient is "under observation" pending further assessment — finalizing the opinion only once full investigation is complete.*

4.2 Legal Documents in Medical Records

> **In easy words:** Apart from clinical notes, a chart also holds several "legal papers" that prove consent was given, the patient's identity is recorded, or the patient refused treatment.

- Consent forms (informed consent for treatment, surgery, procedures).
- MLC register.
- Post-mortem report.
- Wound certificate.
- Birth certificate.
- Death certificate.
- Discharge Against Medical Advice (DAMA) form.
- Referral letters, request for information, court summons responses.

4.3 Patient Rights

> **In easy words:** Patients are not just passive subjects of the record — they have specific rights over their own information and care decisions.

- Right to access own medical records.

- Right to confidentiality of personal health information.
- Right to informed consent before any procedure.
- Right to correction of errors in records.
- Right to refuse treatment (DAMA).

4.4 Confidentiality

> **In easy words:** A patient's information is private and cannot be shared with outsiders unless the patient agrees, a court orders it, or public health law requires it.

- Patient information cannot be disclosed without written consent **except:**
  - To the treating healthcare team (on a need-to-know basis).
  - By court order or legal summons.
  - To public health authorities for notifiable/communicable diseases.
- HIPAA (Health Insurance Portability and Accountability Act): mandates national standards for protecting PHI (Protected Health Information). Covers confidentiality of paper and electronic medical records.
- Medical coders must ensure data is not accessed by unauthorized persons.

Example: *A relative calls the hospital asking for a patient's HIV test result over the phone — the staff must refuse, since disclosure without the patient's written consent (and not to an authorized care team member) would breach confidentiality, regardless of the relationship.*

4.5 Medical Audit

> **In easy words:** A medical audit is a regular health check-up for the hospital's own quality of care — checking records against standards, finding gaps, fixing them, and checking again.

Medical audit is a **systematic review of medical care quality** against established standards, aimed at improving patient outcomes and healthcare quality.

Objectives of Medical Audit:

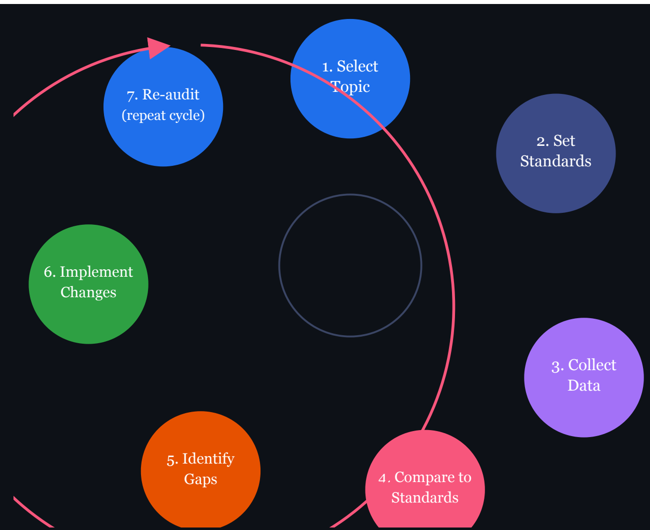
- Improve patient care quality and reduce errors.
- Support accreditation (NABH, JCI).
- Ensure accountability of medical and nursing staff.
- Identify deficiencies in care delivery.
- Formulate health policies and programs.

Types of Medical Audit:

Type	Description
Prospective Audit	Reviews care standards before or during patient care to guide treatment.
Concurrent Audit	Ongoing review during patient's current episode of care.
Retrospective Audit	Review of care quality after the patient has been discharged, based on existing records.
Internal Audit	Conducted by staff within the hospital. MRD staff carry out quantitative analysis; medical personnel carry out qualitative analysis.
External Audit	Conducted by persons outside the treating team — hospital administrator with selected clinicians; medical audit committee; or specialists appointed by Director of Medical & Health Services.
Morbidity Audit	Reviews and records the main condition responsible for the patient's need for treatment or investigation.

The Audit Cycle:

> **In easy words:** The audit cycle never really ends — once you fix a problem, you check again later to make sure the fix actually worked, and the cycle repeats.



Picture 4.2 — The 7-step Audit Cycle, looping back from Re-audit to Select Topic

1. Select Topic → 2. Set Standards → 3. Collect Data → 4. Compare with Standards → 5. Identify Gaps → 6. Implement Changes → 7. Re-audit

(cycle repeats from Step 1)

Two Phases of Medical Auditing:

1. **First Phase (Quantitative):** Providing adequate records of performance as a basis of analysis (ensuring records are complete and available).
2. **Second Phase (Qualitative):** Actual analysis of recorded data — reviewing clinical records and filed reports pertaining to professional work of the hospital.

Internal Audit Process:

- MRD staff collect all records of discharged patients daily and arrange in chronological order.
- Deficiencies of each record listed in a deficiency slip.
- Doctors complete deficient records during weekly doctors' conference / MRD room.
- Medical officers from other units can be asked to countercheck records and verify if treatment was consistent with diagnosis.

External Audit — Methods:

1. Hospital administrator + selected clinicians: monthly medical statistical meetings to discuss previous month's hospital statistics.
2. Medical Audit Committee (5–10 physicians/surgeons): frank, fearless, skilled, without prejudice.
3. Director of Medical and Health Services selects specialists for periodic visits to examine professional work and make confidential reports to the medical superintendent.

Common Deficiencies Found in Audit:

1. Diagnosis without complete description; improper terminology.
2. Detailed diagnosis written inside but not on the face sheet.
3. Different final diagnosis on face sheet vs. discharge summary.
4. Discharge notes incomplete or missing; only "D" written instead of full note.
5. No provisional diagnosis on history and physical examination record.
6. Procedures/operations not recorded on face sheet.
7. When provisional and final diagnosis are the same, written as "same as above" (incorrect).
8. Result column not completed.
9. Signature of head of service/unit not found.
10. Description of accident not written in RTA cases.

11. Usage of improper medical forms.
12. Date and time of examination not written.

Solutions to Minimise Audit Deficiencies:

MRO
Establish written policies, standard forms, and guidelines. Maintain effective incomplete record control. Create doctors' conference room in MRD.

Doctor
Timely documentation immediately after attending patient. Senior doctors verify junior doctors' documentation. Documentation time should be considered clinical time.

Nurse/Paramedical Staff
Complete all related records of their services promptly.

Administrator
Provide transcription service; institute rewards for prompt completion and penalties for delayed completion.

Example: *A retrospective internal audit of last month's discharge summaries finds 15% have no head-of-unit signature. The MRO reports this to administration; the hospital responds by making "documentation completion" part of the consultant's weekly performance review — closing the loop until the next audit cycle confirms improvement.*